# [BLR-927] DWB Manila Water Payment Flow — V1 Baseline Benchmark

BLR Platform runtime for the Payment Link (Dynamic Web Biller) MANILA WATER COMPANY flow · payment gateway (3DS) shown separately · baseline before the code enhancement

| | |
|---|---|
| **Ticket** | BLR-927 · https://apnsolutions.atlassian.net/browse/BLR-927 |
| **Environment** | Test — V1, before the code enhancement |
| **Entry URL** | `https://test-pay.billeroo.com/7227265c-e920-470b-9dec-295dbe216c16/payment` |
| **Measured** | 16 Oct 2026, 11:09–11:12 PHT |
| **Runs averaged** | 4 (all paid and posted successfully) |
| **BLR Platform steps** | 1. Open payment page · 2. Form to Payment Summary · 3. Open consent · 4. Consent to TLPE · 5. TLPE to Confirm Payment · 8. Result to receipt |
| **Test data** | Contract Account 24312673 (reusable), Account Name "Justine Agner", a different random amount per run (biller blocks the same account + amount twice a day) |
| **Excluded (gateway)** | 6. Pay to 3DS · 7. 3DS to Payment Result (Visa test card, 3DS Simulator → Approve) |

**BLR Platform runtime: 20.45 s on average** (14.74–27.57 s across 4 runs), with the payment gateway excluded. The slowest BLR step is **Consent to TLPE** at 6.63 s (32% of BLR time). Including the 3DS payment gateway, the full flow takes 45.32 s, so the gateway accounts for 55% of what a payer waits.

**Read with care: the test environment was slower during this run.** The payment gateway steps are the same TLPE/3DS pages for SWB and DWB, yet they took 24.88 s here vs 12.94 s in the SWB run the day before (PAY → 3DS 8.96 s vs 2.88 s). Run-to-run spread is also wide. Compare this baseline only with an after-enhancement run taken at a similar time, and do not compare it directly with the SWB baseline.

BLR PLATFORM RUNTIME (AVG)
**20.45 s**
Median 19.74 s · gateway excluded

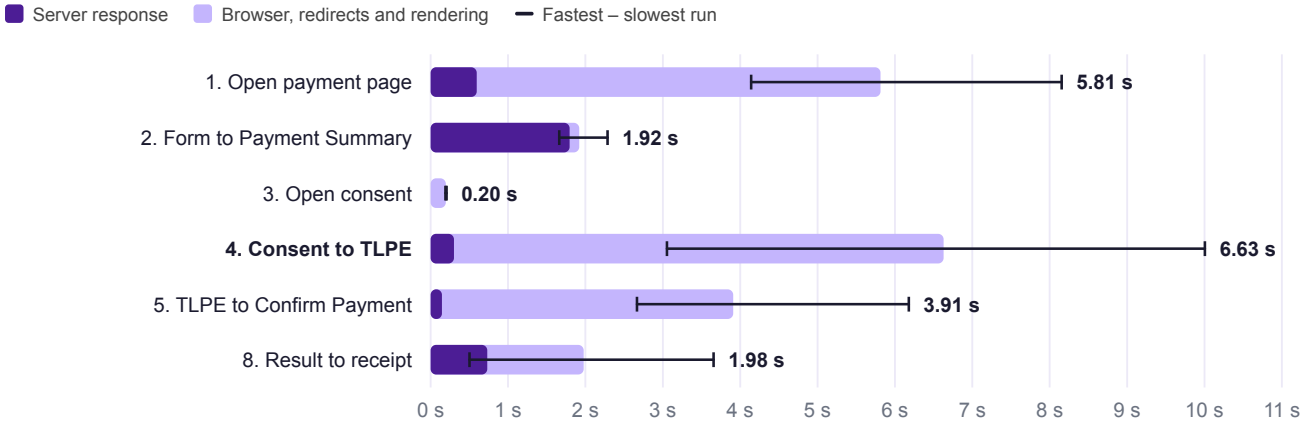
BLR FASTEST – SLOWEST
**14.74–27.57 s**
Spread 12.82 s

SLOWEST BLR STEP
**6.63 s**
Consent to TLPE · 32%

FOR REFERENCE: WITH 3DS
**45.32 s**
+24.88 s payment gateway

## BLR Platform — average time per step BLR ONLY

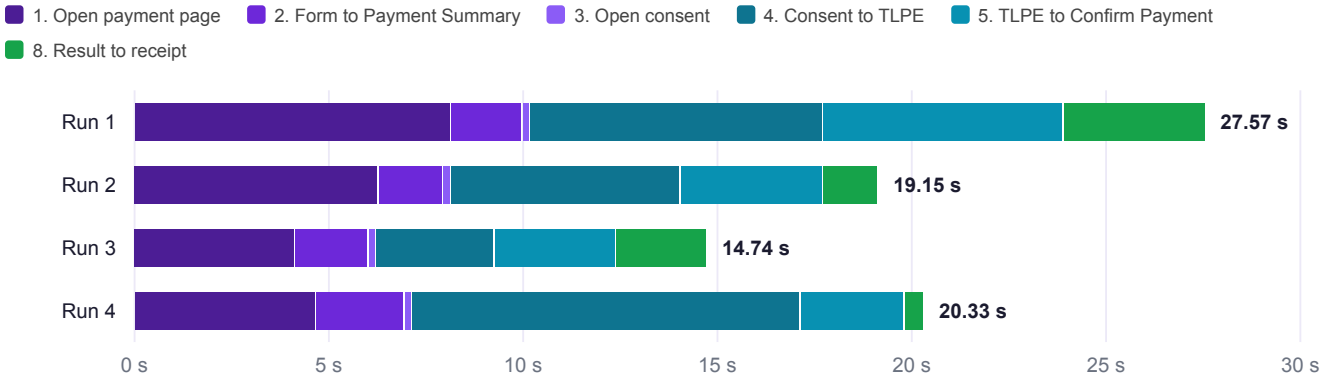
Bar = 4-run average. Dark part = server response (time to first byte); light part = page loading, redirects and rendering. Black line = fastest to slowest run. The slowest step is in bold.



## BLR Platform — runtime per run

BLR ONLY

Each bar is one paid and posted transaction, built from its 6 BLR Platform steps in order.



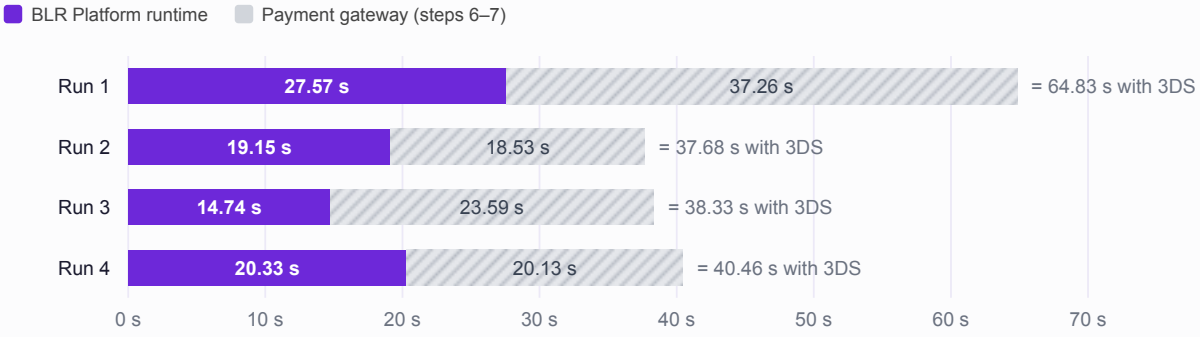
## BLR Platform — step timings in detail (seconds) BLR ONLY

Each step starts at the click and ends when the next page is ready. Typing and form filling are not timed. Server avg is the time to first byte of the page the step loads; Open consent opens a pop-up, so it has none. Share is of BLR Platform runtime.

| # | STEP | RUN 1 | RUN 2 | RUN 3 | RUN 4 | AVERAGE | MIN | MAX | SERVER AVG | SHARE |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | **Open payment page**<br>Load /payment until the form shows | 8.15 | 6.29 | 4.14 | 4.67 | **5.81** | 4.14 | 8.15 | 0.60 | 28% |
| 2 | **Form to Payment Summary**<br>Next until the Payment Summary shows | 1.85 | 1.66 | 1.89 | 2.29 | **1.92** | 1.66 | 2.29 | 1.79 | 9% |
| 3 | **Open consent**<br>Next until the Consent pop-up shows | 0.19 | 0.21 | 0.20 | 0.19 | **0.20** | 0.19 | 0.21 | — | 1% |
| 4 | **Consent to TLPE**<br>Proceed until TLPE Customer Details shows | 7.54 | 5.91 | 3.05 | 10.01 | **6.63** | 3.05 | 10.01 | 0.30 | 32% |
| 5 | **TLPE to Confirm Payment**<br>Next (Visa) until the card form shows | 6.18 | 3.67 | 3.12 | 2.67 | **3.91** | 2.67 | 6.18 | 0.15 | 19% |
| 8 | **Result to receipt**<br>Click "here" until the Payment Posted receipt shows | 3.66 | 1.41 | 2.34 | 0.50 | **1.98** | 0.50 | 3.66 | 0.73 | 10% |
| | **BLR Platform runtime**<br>Steps 1–5 and 8, payment gateway excluded | **27.57** | **19.15** | **14.74** | **20.33** | **20.45** | 14.74 | 27.57 | 3.57 | 100% |

### With vs without the payment gateway (3DS) REFERENCE ONLY

Steps 6 and 7 happen on the payment gateway (TLPE PAY, the 3DS Simulator and its redirects back), not on the BLR Platform, so they are excluded from the BLR runtime above. They are shown here so the full payer wait is still on record.




| RUN | BLR PLATFORM (S) | PAYMENT GATEWAY (S) | FULL FLOW WITH 3DS (S) | GATEWAY SHARE |
|---|---|---|---|---|
| Run 1 | **27.57** | 37.26 | 64.83 | 57% |
| Run 2 | **19.15** | 18.53 | 37.68 | 49% |
| Run 3 | **14.74** | 23.59 | 38.33 | 62% |
| Run 4 | **20.33** | 20.13 | 40.46 | 50% |
| **Average** | **20.45** | **24.88** | **45.32** | **55%** |

## Payment gateway steps in detail (seconds) EXCLUDED FROM BLR

| # | STEP | RUN 1 | RUN 2 | RUN 3 | RUN 4 | AVERAGE | MIN | MAX | SERVER AVG |
|---|---|---|---|---|---|---|---|---|---|
| 6 | **Pay to 3DS**<br>PAY until the 3DS Simulator shows | 12.55 | 4.23 | 11.31 | 7.77 | **8.96** | 4.23 | 12.55 | 0.84 |
| 7 | **3DS to Payment Result**<br>Approve + Submit until TRANSACTION SUCCESSFUL | 24.71 | 14.30 | 12.28 | 12.36 | **15.91** | 12.28 | 24.71 | 0.18 |
| | **Payment gateway time**<br>Steps 6–7 (TLPE PAY, 3DS and its redirects) | **37.26** | **18.53** | **23.59** | **20.13** | **24.88** | 18.53 | 37.26 | — |

Most of step 7 is browser redirects after 3DS (oppwa → TLPE callback → status page); its server time is only 0.18 s.

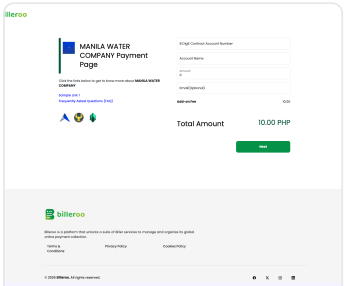
## The 4 benchmark transactions

| RUN | PAID AT (PHT) | ACCOUNT | TRANSACTION REF | BILL (PHP) | ADD-ON (PHP) | BLR RUNTIME (S) | WITH 3DS (S) | RECEIPT |
|---|---|---|---|---|---|---|---|---|
| Run 1 | 2026-10-16 11:09:25 | 24312673 | 261016030925NNN2766 | 32.90 | 10.00 | **27.57** | 64.83 | ● Payment has been posted successfully |
| Run 2 | 2026-10-16 11:10:30 | 24312673 | 261016031030HNH1004 | 50.34 | 10.00 | **19.15** | 37.68 | ● Payment has been posted successfully |
| Run 3 | 2026-10-16 11:11:05 | 24312673 | 261016031105MBZ0566 | 77.94 | 10.00 | **14.74** | 38.33 | ● Payment has been posted successfully |
| Run 4 | 2026-10-16 11:11:56 | 24312673 | 261016031156IJQ1413 | 81.82 | 10.00 | **20.33** | 40.46 | ● Payment has been posted successfully |

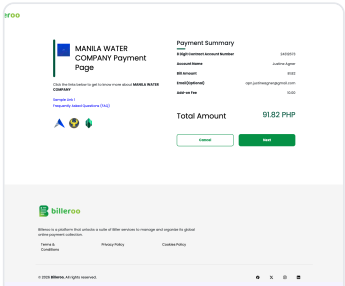
The receipt also adds a card Service Fee from TLPE (not shown on the DWB Payment Summary), so the payer is charged bill + add-on + service fee. Paid-at time is taken from the transaction reference.

## Evidence — Run 4 (txn 261016031156IJQ1413, PHP 81.82)

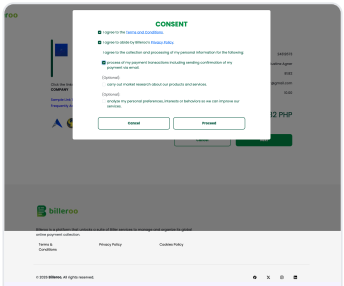
One screenshot per step, captured by the benchmark. Dashed frames are payment gateway pages. Every run is also recorded on video in `benchmarks/dwb-manila-water/videos/`.



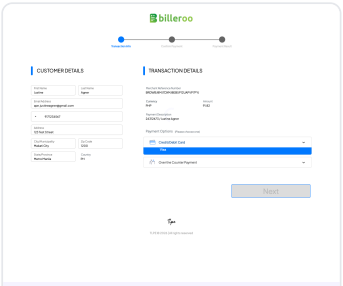

1. Payment page



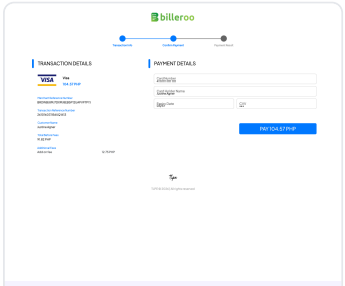

2. Payment Summary



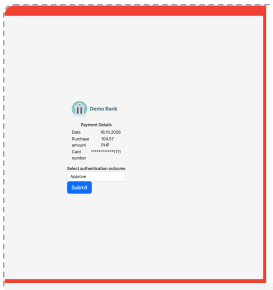

3. Consent



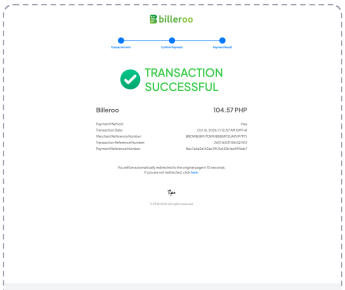

4. TLPE Customer Details + Visa



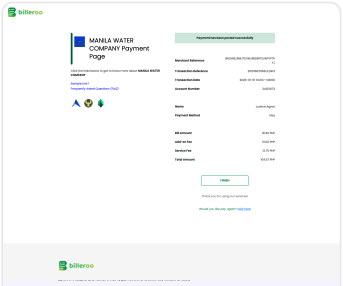

5. TLPE Confirm Payment




6. 3DS Simulator (gateway)




7. TRANSACTION SUCCESSFUL (gateway)




8. Payment Posted receipt

## Method and notes

- **BLR Platform runtime** = steps 1–5 and 8: payment page, form to Payment Summary, Consent, hand-off to TLPE Customer Details, TLPE Next to Confirm Payment, and the return to the Payment Posted receipt. Same split as the SWB baseline.
- **Payment gateway (excluded)** = steps 6–7: PAY → 3DS Simulator, and Approve + Submit → TRANSACTION SUCCESSFUL.
- **Fresh visitor each run:** new browser session with no cookies or cache.
- **Headed browser required:** in headless Chromium, TLPE shows "Sorry, something went wrong" after PAY and never reaches 3DS.
- **Only posted runs count:** if the receipt says Payment Posting Failed, the run repeats with a new amount. All 4 runs posted on the first attempt.
- **Possible bug found:** clicking Next on the Payment Summary before the page finishes loading skips the Consent pop-up and goes straight to TLPE (same as SWB: the Consent handler is only attached once the page script has run).
- **After the code enhancement:** run `BENCH_LABEL=V1-enhanced BENCH_RUNS=4` with `manila-water.benchmark.spec.ts` at a similar time of day and compare the BLR Platform runtime against this baseline.